SE444: Introduction to Artificial Intelligence
Final Exam — Fall 2025

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Date: Monday, December 22, 2025 Time: 12:30 PM – 2:30 PM (120 min)

Instructions

- This is a closed-book exam. The University's code of ethics applies.
- Answer each exam question to the best of your knowledge.
- Show your work – any sign of serious effort will be considered.

Student Information

Student ID:	Student Name:

Summary of Exam Questions

Part	Topic	Time	Score	Mark	CLOs
I MCQ	BN Structure / d-Sep	5m	1		1
	Bayes' Rule	5m	1		1
	Decision Theory / MDPs	5m	1		1,2
	Game Theory	5m	1		1,3
	Planning (STRIPS)	5m	1		1,2
	Project Code	5m	1		2
II Robot Nav	BN Robot Perception	15m	4		1
	Multi-Sensor Fusion	15m	4		1,2
	Path Decision (MEU)	15m	5		1,2
	Project Code (Bayes)	15m	3		2
III Multi-Robot	STRIPS Planning	12m	3		1,2
	Coordination Game	10m	3		1,3
	Project: Logic Agent	8m	2		2
	TOTAL	120m	**30**		

CLOs: 1 = Knowledge & Understanding, 2 = Skills (Implementation), 3 = Values & Ethics

Part I. Critical-Thinking Multiple Choice Questions

Total Score: 6 points **Total Time:** 30 minutes

Instructions: Each question may have **one or more correct answers**. Select **all that apply**. Partial credit will be awarded for partially correct answers.

Question I.1: Bayesian Network Structure & d-Separation

Time: 5 min **Score:** 1 pt **CLOs:** CLO 1

Consider the Bayesian Network: $A \to C \leftarrow B$, $A \to D \leftarrow E$, $C \to F$

Which conditional independence statement is TRUE?

A) $A \perp B \mid C$

B) $A \perp E \mid D$

C) $A \perp B \mid F$

D) $B \perp E$

E) $A \perp F$

Question I.2: Bayes' Rule Application

Time: 5 min **Score:** 1 pt **CLOs:** CLO 1

A forensic DNA analysis in a criminal investigation: $P(\text{Guilty}) = 0.005$ (1 in 200 suspects), $P(\text{Match}|\text{Guilty}) = 0.99$, $P(\text{Match}|\neg\text{Guilty}) = 0.03$ (false positive). A DNA match is found. Which statement is TRUE?

A) $P(\text{Guilty}|\text{Match}) > 0.5$

B) $P(\text{Guilty}|\text{Match})$ is between 0.10 and 0.20

C) $P(\text{Guilty}|\text{Match}) < 0.05$

D) If $P(\text{Guilty})$ increases to 0.20, then $P(\text{Guilty}|\text{Match}) > 0.95$

E) $P(\text{Guilty}|\text{Match}) < P(\text{Match}|\text{Guilty})$

Question I.3: Decision Theory & MDPs

Time: 5 min **Score:** 1 pt **CLOs:** CLO 1, 2

An autonomous robot uses an MDP with discount factor $\gamma = 0.8$ for navigation. At state s, the robot has two actions:

- **Action A:** Immediate reward $R = 10$, transitions to a terminal state
- **Action B:** Immediate reward $R = 2$, transitions to state s' where $V(s') = 15$

Which statement is TRUE?

A) Action A is optimal at state s

B) $V(s) = 10$ if the robot always chooses Action A

C) $V(s) = 10$ under the optimal policy

D) $Q(s, A) > Q(s, B)$

E) Changing γ to 0.5 would not affect which action is optimal

Question I.4: Game Theory — Nash Equilibrium

Time: 5 min **Score:** 1 pt **CLOs:** CLO 1, 3

Consider the game (payoffs Row, Col):

	Left	**Right**
Up	(2, 1)	(1, 2)
Down	(3, 0)	(0, 3)

Which statement is TRUE?

A) Row player has a strictly dominant strategy

B) (Up, Left) is a Nash Equilibrium

C) (Down, Right) is a Nash Equilibrium

D) (Down, Left) is a Nash Equilibrium

E) Column player has a strictly dominant strategy

Question I.5: Classical Planning — STRIPS

Time: 5 min **Score:** 1 pt **CLOs:** CLO 1, 2

Consider the following STRIPS action schema for UAV navigation:

```
Action: FlyTo(uav, from, to)
  Precond: At(uav, from), HasBattery(uav), Cleared(to)
  Add:     At(uav, to), Visited(to)
  Delete:  At(uav, from), Cleared(to)
```

Given initial state: `At(UAV1, Base), HasBattery(UAV1), Cleared(WP1), Cleared(WP2)`
After executing `FlyTo(UAV1, Base, WP1)`, which statement is TRUE?

A) `At(UAV1, Base)` is still true

B) `Visited(WP1)` is true

C) `Cleared(WP1)` is still true

D) `HasBattery(UAV1)` becomes false

E) The action cannot execute because `Visited(WP1)` is not in preconditions

Question I.6: Project Code — Logic Agent

Time: 5 min **Score:** 1 pt **CLOs:** CLO 2

```
def get_action(self, percept):
    self.kb.tell(make_percept_sentence(percept, self.time))
    action = None
    for a in self.possible_actions:
        if self.kb.ask(is_safe_action(a, self.time)):
            action = a
            break
    self.kb.tell(make_action_sentence(action, self.time))
    self.time += 1
    return action
```

Which statement is TRUE about this agent code?

A) This is a model-based reflex agent

B) This is a knowledge-based (logic) agent

C) `kb.ask()` modifies the knowledge base

D) The agent uses probability distributions to select actions

E) `self.time` only tracks visited grid cells

Part II. Robot Navigation Under Uncertainty

Total Score: 16 points **Total Time:** 60 minutes

Question II.1: BN for Robot Perception (4 pts, 15 min)

Scenario: A navigation robot uses multiple sensors to detect obstacles:

- `Obstacle (O)`: True obstacle present at location
- `Camera (C)`: Camera detects obstacle
- `Lidar (L)`: Lidar detects obstacle
- `Weather (W)`: Bad weather conditions
- `PathBlocked (P)`: Robot path is blocked
- `Alert (A)`: System generates navigation alert

Dependencies: O causes both C and L readings. W affects C accuracy. P depends on O and W. A depends on P.

Tasks:

(a) [2 pts] Draw the Bayesian Network structure with all 6 nodes.

(b) [1 pt] Are $L \perp A \mid P$? Use d-separation. Trace all paths.

(c) [1 pt] Identify ALL collider nodes and explain "explaining away" for one.

Question II.2: Multi-Sensor Fusion (4 pts, 15 min)

Scenario: A robot uses TWO sensors sequentially to update its belief about an obstacle.

Prior: $P(\text{Obstacle} = T) = 0.2$

Sensor Accuracies:

- **Camera:** $P(\text{Detect}|\text{Obstacle} = T) = 0.8$, $P(\text{Detect}|\text{Obstacle} = F) = 0.15$
- **Lidar:** $P(\text{Detect}|\text{Obstacle} = T) = 0.9$, $P(\text{Detect}|\text{Obstacle} = F) = 0.1$

Observations: Camera detects obstacle (1st), then Lidar detects obstacle (2nd).

Tasks:

(a) [2 pts] After Camera=Detect, compute $P(\text{Obstacle} = T \mid \text{Camera} = \text{Detect})$. This becomes the new prior. *Write one sentence interpreting the result.*

(b) [2 pts] After Lidar=Detect, compute $P(\text{Obstacle} = T \mid \text{Camera} = \text{Detect}, \text{Lidar} = \text{Detect})$ using the posterior from (a) as the prior. *Write one sentence interpreting the result.*

Question II.3: Sensor-Based Navigation Decision (5 pts, 15 min)

Scenario: A robot uses a proximity sensor before deciding to cross a corridor. The sensor reading depends on whether an obstacle is actually present.

Prior: $P(\text{Obstacle} = T) = 0.3$

Sensor CPT:

Sensor Reading	Obstacle=T	Obstacle=F
Alert	0.85	0.10
Clear	0.15	0.90

Utility Table (robot's payoff):

Action	Obstacle=T	Obstacle=F
Cross	-50	$+80$
Wait	$+10$	$+10$

Tasks:

(a) [2 pts] The sensor reads **Alert**. Compute $P(\text{Obstacle} = T \mid \text{Sensor} = \text{Alert})$. Show all steps.

(b) [3 pts] Given Sensor=Alert, compute the **Expected Utility (EU)** for each action and determine the **MEU-optimal** action. Explain why this action is rational.

Question II.4: Project Code Analysis — Bayesian Reasoning (3 pts, 15 min)

Given: The following code from the RoboMind project's `bayes_reasoning.py`:

```
def bayes_update(prior, likelihood, evidence):
    """
    Update belief using Bayes' Rule.
    P(H|E) = P(E|H) * P(H) / P(E)
    """
    if evidence == 0:
        return prior  # Avoid division by zero
    posterior = (likelihood * prior) / evidence
    return posterior

def compute_evidence(prior, likelihood_h, likelihood_not_h):
    """
    P(E) = P(E|H)*P(H) + P(E|~H)*P(~H)
    """
    return (likelihood_h * prior) + (likelihood_not_h * (1 - prior))
```

Scenario: The robot's sensor has the following characteristics:

- $P(\text{Detect} \mid \text{Obstacle}) = 0.9$ (True Positive Rate)
- $P(\text{Detect} \mid \neg\text{Obstacle}) = 0.15$ (False Positive Rate)

The robot's prior belief that a cell contains an obstacle is $P(\text{Obstacle}) = 0.4$.

Tasks:

(a) [1 pt] The sensor reports **Detect**. Trace through the `compute_evidence()` function call. Show the values of each variable and the return value.

(b) [1 pt] Trace through the `bayes_update()` function to compute the posterior probability. What is $P(\text{Obstacle} \mid \text{Detect})$?

(c) [1 pt] If the code used `evidence = 0.5` instead of computing it properly, what would the posterior be? Why is computing evidence correctly important?

Part III. Multi-Robot Cooperation

Total Score: 8 points **Total Time:** 30 minutes

Question III.1: Robot Task Planning — STRIPS (3 pts, 12 min)

Rescue Robot Scenario: A rescue robot must pick up a victim and deliver them to a safe zone.
Predicates: `At(robot,loc)`, `Holding(robot,victim)`, `VictimAt(victim,loc)`, `Safe(loc)`
Initial State: At(R1, Base), VictimAt(V1, Zone3), Safe(Zone1), ¬Holding(R1, V1)
Goal State: VictimAt(V1, Zone1)
Tasks:

(a) [1 pt] Define STRIPS action `Pickup(robot, victim, loc)`: Preconditions, Add List, Delete List.

(b) [2 pts] Show a forward search plan to achieve the goal. Include actions: MoveTo, Pickup, PutDown.

Question III.2: Robot Coordination Game (3 pts, 10 min)

Scenario: Two delivery robots (R1 and R2) approach an intersection. Each must choose a lane (Left or Right) to avoid collision.

R1 \ R2	Left	Right
Left	(0, 0)	(5, 5)
Right	(5, 5)	(0, 0)

Tasks:

(a) [1 pt] Identify any strictly dominated strategies.

(b) [2 pts] Find all pure-strategy Nash Equilibria. Mark best responses.

Question III.3: Project Code — Logic Agent (2 pts, 8 min)

Given: Knowledge Base code from the RoboMind project's `logic_agent.py`:

```
def get_safe_moves(self, percept):
    # Add current sensor readings to KB
    for direction in ["North", "East", "South", "West"]:
        if percept[direction] == "wall":
            self.kb.tell(f"Wall({direction})")
        else:
            self.kb.tell(f"Free({direction})")

    # Apply rule: Free(X) => Safe(X)
    self.kb.infer()

    # Find all safe moves
    safe_moves = []
    for action in ["North", "East", "South", "West"]:
```

```
        if self.kb.ask(f"Safe({action})"):
            safe_moves.append(action)

    return safe_moves
```

Scenario: The robot's sensors detect: wall to North, wall to East, free to South, free to West.
Tasks:

(a) [1 pt] What is the purpose of `kb.tell()` vs `kb.ask()`? How do they differ?

(b) [1 pt] Trace through the for-loop that builds `safe_moves`. For each direction, show whether `kb.ask()` returns True or False, and list the final contents of `safe_moves`.

Reference Sheet — Formulas & Concepts

Bayesian Inference

Bayes' Rule:

$$P(H \mid E) = \frac{P(E \mid H) \cdot P(H)}{P(E)}$$

Total Probability:

$$P(E) = \sum_i P(E \mid H_i) \cdot P(H_i)$$

For Binary Hypothesis:

$$P(E) = P(E|H)P(H) + P(E|\neg H)P(\neg H)$$

MDPs & Value Functions

Q-Value (Action-Value):

$$Q(s,a) = R(s,a) + \gamma \sum_{s'} P(s' \mid s,a) V(s')$$

Deterministic Transition:

$$Q(s,a) = R(s,a) + \gamma \cdot V(s')$$

Value Function:

$$V(s) = \max_a Q(s,a)$$

Optimal Policy:

$$\pi^*(s) = \arg\max_a Q(s,a)$$

Decision Under Uncertainty

Expected Utility:

$$EU(a) = \sum_s P(s) \cdot U(a,s)$$

Maximum Expected Utility:

$$a^* = \arg\max_a EU(a)$$

Game Theory

Dominant Strategy: A strategy that is always best regardless of opponent's choice.

Nash Equilibrium: A strategy profile where no player can improve by unilaterally changing strategy.

Best Response: Given opponent's strategy, the action that maximizes own payoff.

STRIPS Planning

Action Schema:

```
Action(args):
  Precond: conditions to execute
  Add:     facts that become true
  Delete:  facts that become false
```

Result: $\text{State}' = (\text{State} - \text{Delete}) \cup \text{Add}$

Knowledge-Based Agents

Tell: Add knowledge to KB

Ask: Query KB via inference (read-only)

Infer: Apply rules to derive new facts

This reference sheet provides general formulas. Apply them carefully to specific problem contexts.